

Introduction to Mathematics and Optimisation

Exercises Week 5

Kirsten Pleskot (202507620)

20. 01. 2026

1 Exercise 4.37

A vitamin pill P is produced using two ingredients M_1 and M_2 . The pill needs to satisfy four constraints for the vital vitamins V_1 and V_2 . It must contain at least 6 milligrams and at most 15 milligrams of V_1 and at least 5 milligrams and at most 12 milligrams of V_2 . The ingredient M_1 contains 3 milligrams of V_1 and 2 milligrams of V_2 per gram. The ingredient M_2 contains 2 milligrams of V_1 and 3 milligrams of V_2 per gram:

	V_1	V_2
M_1	3	2
M_2	2	3

Let x denote the amount of M_1 and y the amount of M_2 (measured in grams) in the production of a vitamin pill. Write down a system of linear inequalities in x and y describing the constraints above.

We want a vitamin pill of minimal weight satisfying the constraints. How many grams of M_1 and M_2 should we mix?

Use Fourier–Motzkin elimination to solve this problem.

Check your solution by modifying the input to the Sage code in Example 4.34 using Remark 4.7.

One may also force minimization by inserting the following option:

```
LP = MixedIntegerLinearProgram(maximization=False, solver="GLPK").
```

1.1 Solution

From the description above we receive the following inequalities:

$$6 \leq 3x + 2y \leq 15$$

$$5 \leq 2x + 3y \leq 12$$

$$x \geq 0$$

$$y \geq 0$$

Which is equivalent to:

$$-3x - 2y \leq -6$$

$$3x + 2y \leq 15$$

$$-2x - 3y \leq -5$$

$$2x + 3y \leq 12$$

$$-x \leq 0$$

$$-y \leq 0$$

And we are trying to minimize $x + y$. We define $z = x + y$ and substitute x with $x = z - y$ to the inequities.

$$-3(z - y) - 2y \leq -6 \tag{1}$$

$$3(z - y) + 2y \leq 15 \tag{2}$$

$$-2(z - y) - 3y \leq -5 \tag{3}$$

$$2(z - y) + 3y \leq 12 \tag{4}$$

$$-(z + y) \leq 0 \tag{5}$$

$$-y \leq 0 \tag{6}$$

After simplifying we can isolate y like this:

$$y \leq 3z - 6 \tag{7}$$

$$3z - 15 \leq y \tag{8}$$

$$-2z + 5 \leq y \tag{9}$$

$$y \leq -2z + 12 \tag{10}$$

$$y \leq z \tag{11}$$

$$0 \leq y \tag{12}$$

Therefore

$$\max(3z - 15, -2z + 5, 0) \leq y \leq \min(3z - 6, -2z + 12, z) \tag{13}$$

We can now eliminate y like this:

$$3z - 15 \leq 3z - 6 \quad (14)$$

$$3z - 15 \leq -2z + 12 \quad (15)$$

$$3z - 15 \leq z \quad (16)$$

$$-2z + 5 \leq 3z - 6 \quad (17)$$

$$-2z + 5 \leq -2z + 12 \quad (18)$$

$$-2z + 5 \leq z \quad (19)$$

$$0 \leq 3z - 6 \quad (20)$$

$$0 \leq -2z + 12 \quad (21)$$

$$0 \leq z \quad (22)$$

We can now isolate y like this:

$$z \leq \frac{27}{5} \quad (23)$$

$$z \leq \frac{15}{2} \quad (24)$$

$$\frac{11}{5} \leq z \quad (25)$$

$$\frac{5}{3} \leq z \quad (26)$$

$$z \leq 6 \quad (27)$$

$$0 \leq z \quad (28)$$

and from this we get that

$$2.2 \leq z \leq 6 \quad (29)$$

and since we are trying to minimize z

$$z = 2.2 \quad (30)$$

now we can substitute z to (13) and we get

$$0.6 \leq y \leq 0.6 \quad (31)$$

therefore

$$y = 0.6 \quad (32)$$

and since $x = z - y$

$$x = 1.6 \quad (33)$$

1.2 Sage

```
LP = MixedIntegerLinearProgram(solver="GLPK", maximization=False)
v = LP.new_variable(integer=False, nonnegative=True)
x, y = v[0], v[1]

LP.set_objective(x + y)

LP.add_constraint(-3*x - 2*y <= -6)
LP.add_constraint( 3*x + 2*y <= 15)
LP.add_constraint(-2*x - 3*y <= -5)
LP.add_constraint( 2*x + 3*y <= 12)

LP.show()

oval = LP.solve()
print('Optimal Value:', oval)
print('x =', LP.get_values(x))
print('y =', LP.get_values(y))
```

The code shown above gives the output

Minimization:

$x_0 + x_1$

Constraints:

$-3.0 x_0 - 2.0 x_1 \leq -6.0$

$3.0 x_0 + 2.0 x_1 \leq 15.0$

$-2.0 x_0 - 3.0 x_1 \leq -5.0$

$2.0 x_0 + 3.0 x_1 \leq 12.0$

Variables:

x_0 is a continuous variable (min=0.0, max=+oo)

x_1 is a continuous variable (min=0.0, max=+oo)

Optimal Value: 2.1999999999999997

$x = 1.5999999999999999$

$y = 0.6$

Any complaints shall be directed here